

Experimental Methods in Engineering Mechanics: DIC and Crack tip fields

Thomas Lepère,^{*} Paul Nauche,[†] and Theophile de Tournemire[‡]
STI - Mechanical Engineering, École Polytechnique Fédérale de Lausanne (EPFL)
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This study investigates the application of Digital Image Correlation (DIC) for analyzing deformation fields near crack tips in polyacrylamide hydrogels, which are widely used in medical applications due to their biocompatibility. Using a high-contrast random speckle pattern and tracking marker displacements, DIC provides spatially resolved measurements of strain tensors. The research evaluates stress intensity factors for both linear elastic (K_{linear}) and plastic ($K_{plastic}$) behaviors using experimental data, including the J-integral method. Results show that the linear elastic assumption aligns well with experimental observations for most cases. The study highlights the limitations of traditional fracture mechanics in soft materials exhibiting nonlinear behavior and explores how experimental methods can address these challenges.

Keywords: J-integral, Linear intensity stress factor, Plastic intensity stress factor, Polyacrylamide hydrogel, DIC, Tensile test, Neo-Hookean

Introduction.—Fracture mechanics plays a crucial role in understanding the failure mechanisms of materials under stress, particularly when assessing the behavior of cracks. To study these mechanics and evaluate the material properties, we apply Digital Image Correlation (DIC), an advanced image-based technique, to investigate the deformation fields near crack tips. DIC enables the tracking of a high-contrast random pattern applied to the surface of a specimen, providing detailed spatial information about the deformation by comparing the displacement of markers between a reference and deformed state. This technique allows for the determination of key kinematic quantities, such as strain tensors, which are critical for understanding material behavior.

A central parameter in assessing the fracture behavior of a material in linear elastic fracture mechanics is the stress intensity factor K_{linear} , expressed as

$$K_{linear} = \sigma Y \sqrt{\pi a} \quad (1)$$

σ is the applied nominal stress, Y is a geometry-dependent factor, and a is the crack length. This formulation assumes small deformations and elastic material behavior. However, many soft materials, such as hydrogels, exhibit nonlinear, large deformations that challenge the assumptions of linear elasticity and may cause traditional fracture mechanics models to break down. How does the conventional linear stress intensity factor (K) apply to hydrogels under larger plastic deformations?

Material and methods.—To assess this question, we first review the measurement setup, followed by the calculation of the plastic deformation stress intensity factor,

$K_{plastic}$, using the J-integral. We next compute the stress intensity factor under linear assumptions, K_{linear} . Finally, we compare the results and present the conclusions regarding the assumptions of linear elastic behavior holding under the tested conditions.

We are focusing on polyacrylamide hydrogels with a cross linker-to-monomer molar ratio $C = 4.33 \cdot 10^{-2}$ and a water-to-monomer molar ratio $W = 24.06$. The fracture process was recorded and later analyzed using Digital Image Correlation (DIC). It is an incompressible Neo-Hookean (Wang *et al.* [1]) material with a Young modulus of $E = (179 \pm 22)$ kPa (Lepère *et al.* [2]) and a shear modulus of :

$$\mu = \frac{E}{2(1+\nu)} \quad (2)$$

For an incompressible material, $\nu = 0.5$ which reduce the equation of the shear modulus (2) to $\mu = \frac{E}{3}$. The study is in a plane stress assumption with only one component force acting on the sample and a negligible thickness T compared to the width W .

The video frames of the samples are analyzed using Digital Image Correlation (DIC), a technique employed to measure displacement and strain fields within a designated region of interest (ROI) in materials undergoing deformation. The method involves tracking specific subsets (patterns) from a reference image (representing the undeformed state) and identifying their corresponding locations in the deformed configuration, thereby enabling the determination of the displacement field. In this study, the DIC analysis is performed using an open-source toolbox, Ncorr ([3]), which facilitated the extraction of displacement fields for subsequent computational analyses.

The camera used for the DIC is the *2.1 MP sCMOS* [4] with a resolution of 1920x1080 pixels and the *R3L1S4P* from Thorlabs [5] for the calibration.

To prepare the samples for the DIC, they were spray-painted (Fig. 1) using a specific method detailed in an

^{*} thomas.lepere@epfl.ch

[†] paul.nauche@epfl.ch

[‡] theophile.detournemire@epfl.ch

EPFL course document from *Experimental Methods in Mechanical Engineering*[6]. This method ensures a homogeneous speckle pattern suitable for DIC analysis.

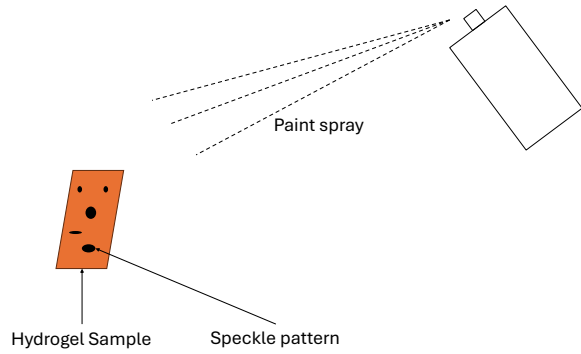


FIG. 1: Schematic for the painting method of the sample. The canister is oriented downwards, and the wind deposits the finest paint particle on the sample.

The experiment is made on a total of 3 samples named "A", "B" and "C". The tensile machine (Fig. 2) moves at a speed of $20 \cdot 10^{-6} \text{ m} \cdot \text{s}^{-1}$ and the camera takes pictures at a frequency of 10 Hz.

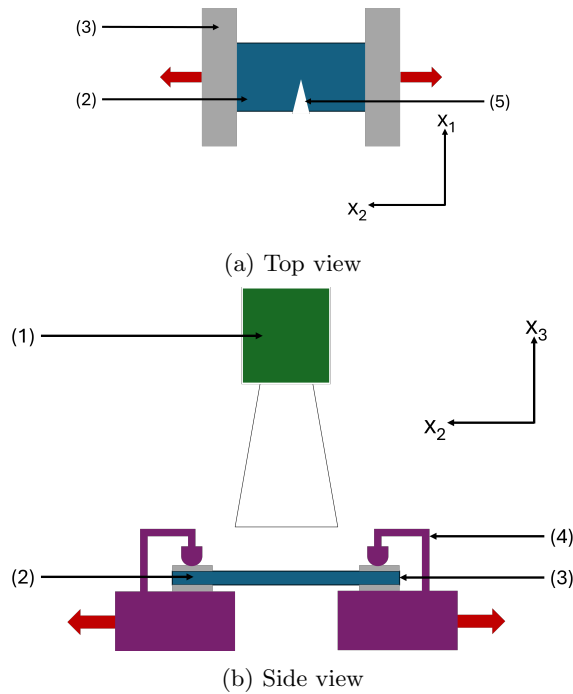


FIG. 2: The tensile test is proceeded as follow: the sample(2) is placed between two plates(3) and is clamped on both ends using magnets(4) and gently pre-stretched. A crack(5) is introduced at one edge using scissors. The camera(1) is focused, and an initial reference image was taken for DIC analysis then a force is applied (red arrows) in a single component direction

Theory : Calculation of plastic K .—The data returned by Ncorr is the displacement field $\mathbf{y} = (u, v)$. Hence, the deformation gradient tensor can be computed using the formula 3. For an incompressible material, $\det(\mathbf{F}) = 1$.

$$\mathbf{F} = \nabla \mathbf{y} + \mathbf{I} \quad (3)$$

For a Neo-Hookean material, the first Piola-Kirchhoff stress tensor is used to express the stress relative to reference configuration. It is expressed as :

$$\mathbf{P} = \mu \mathbf{F} - p \mathbf{F}^{-T} \quad (4)$$

with μ , the shear modulus of the material and

$$p = \mu \cdot \lambda^2 = \frac{E \cdot (\det(F_{ij}))^{-2}}{3} \quad i, j = 1, 2 \quad (5)$$

, the hydrostatic pressure that ensure the incompressibility of our material.

When studying large deformations in the elastic-plastic regime, the displacement field is used to compute the J -integral in equation (8) (Rice, 1968 [7], Liu and Moran, 2020 [8]), serving as a measure of the energy release rate per unit fracture area. Using the following relation, we obtain an experimental value for the stress intensity factor in plastic regime :

$$K_{\text{plast}} = \sqrt{\tilde{E} \cdot J_1} \quad (6)$$

where J_1 is the J -integral, $\tilde{E}$ the effective Young Modulus defined as

$$\tilde{E} = \begin{cases} \frac{E}{1-\nu^2} & \text{plane strain} \\ E & \text{plane stress} \end{cases} \quad (7)$$

, and ν is the Poisson's ratio.

The J -integral is expressed in Equation 8. An example path is showed on Figure 3, the path must be surrounding the crack tip otherwise it is null.

$$J_1 = \int_{\Gamma} (W \delta_{1j} - P_{ij} \cdot y_{i,1}) n_j d\Gamma, \quad i, j = 1, 2 \quad (8)$$

where $\mathbf{y} = (u, v)$ the x_1 and x_2 displacement field. And, W is the strain energy density :

$$W = \frac{1}{2} \mu (I_1 - 2) \quad (9)$$

With, I_1 , the first invariant of the Cauchy stress tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$

$$I_1 = \text{tr}(\mathbf{F}^T \mathbf{F}) \quad (10)$$

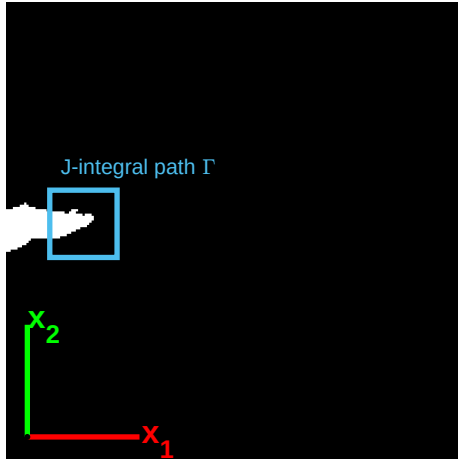


FIG. 3: Schematic of the region of interest (in black) of the DIC. The system of coordinates used is the same as the one used for the J-integral (eq. 8) computation which path is visible.

Theory: Calculation of linear K. — The stress intensity factor K in linear assumption is given by :

$$K_{linear} = 1.12\sigma\sqrt{a\pi} \quad (11)$$

The crack is located on the edge of the sample for planar stress field and the fracture is in mode I (Fig. 4). Therefore, the geometric correction factor is equal to 1.12 [9]. The nominal stress σ in the sample is measured with the force F and the cross section area A_C : $\sigma = \frac{F}{A_C}$. The cross-section area of the sample has been measured, for many samples, with an electronic caliper precise up to 10^{-5} meter (Width x Thickness) and the values are reported in Table III.

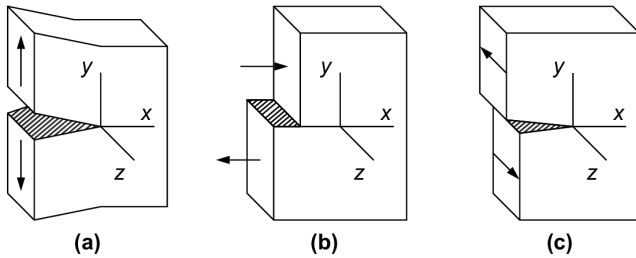


FIG. 4: Schematic representation of faulting modes: (a) Mode I : Opening, characterized by vertical displacement due to extensional forces; (b) Mode II: shearing, involving lateral displacement along a horizontal plane due to shear forces; and (c) Mode III: Tearing, exhibiting vertical displacement due to compressional forces. The shaded planes indicate the fault surfaces.

Source : *Fracture Mechanics* - Wikipedia

On the other hand, the instantaneous forces F exerted on the sample before propagation of the crack are

recorded throughout the test using a load cell. However this method is susceptible to external noise, which leads us to perform a linear regression to retrieve a final force value. This is based on the linearity assumption we made to compute K_{linear} . Additionally, an offset is present in the values, as the force is not zero before any elongation occurs and after the complete fracture.

The linear regression is applied solely to the data following the onset of elongation up until the point of crack propagation. The start of the traction is identified based on the distance of the two plates holding the sample. The beginning of crack propagation is detected on the plot Force over Time (Fig.8) when the force drops suddenly. The offset value of the load cell has been removed as well. The offset was determined by the mean force value when no forces was applied on the sample.

Lastly, the size of the initial crack is measured with a precise photo and a Resolution Test Targets that helps converting pixels to mm. The one we used is the *R3L1S4P* from Thorlabs [5]. The crack length is measured numerically using the *Fiji* software by taking the shortest distance between the end of the notch and the edge of the sample (Fig. 5).

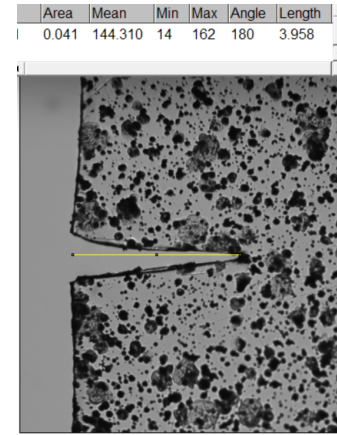


FIG. 5: Screenshot of the software *Fiji* used to measure the size of sample A's crack. The Resolution Test Target has been used to calibrate the software. The measured length is $a = 3.958$ mm.

Presentation of the results.—The displacement field of the sample is determined using the DIC analysis, NOCRR, explained earlier. Once the displacement field is determined, the deformation gradient $\mathbf{F}$ (Eq 3) is deduced. This leads to the calculation of the first Piola-Kirchhoff stress tensor $\mathbf{P}$ (Eq 4).

The J -integral is evaluated around the crack-tip using a counter-clockwise rectangular path (Fig. 3). The value of J is not path dependent but is null when the path is too far from the crack-tip. The evaluation of J is the last step to obtain the value of the stress intensity factor $K_{plastic}$ that we are interested in. The computed values

for J and $K_{plastic}$ are summarized in the Table I.

P_{22} of the Piola-Kirchoff stress tensor

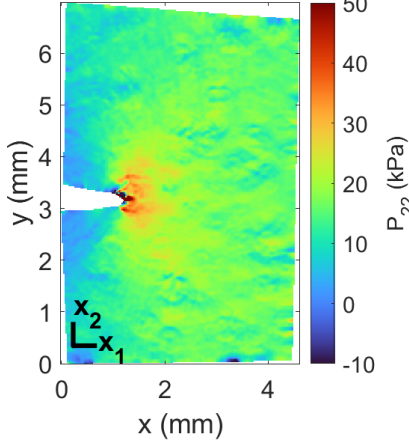


FIG. 6: Second component of the first Piola-Kirchoff stress tensor for the sample A. A high concentration of stress at the crack-tip can be observed. Only the P_{22} component is visualized because it corresponds to the direction of the force exerted on the sample.

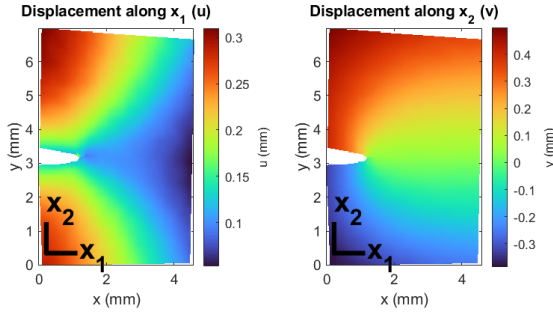


FIG. 7: u displacements (a) and v displacements (b) for the sample A. Given by ncorr with a subset radius of 38, a subset spacing of 3, and the *High Strain Analysis* option with seed propagation enabled.

Sample	A	B	C
$J_1 (\text{J} \cdot \text{m}^{-2})$	16.23	1.61	9.68
$K_{plast} (\text{N} \cdot \text{m}^{3/2})$	1704	536	1317

TABLE I: J-integral and intensity stress factor for each sample in the plastic case.

In the linear assumption, the nominal stress is computed with the cross-section area and the nominal pulling force exerted on the sample before the crack propagation. These are measured as explained in the *Theory* section.

The pulling force raw data exerted on the sample A are shown in the figure below (Fig.8). The force plots for sample B and C can be found in the appendix (Fig.9 & Fig.10).

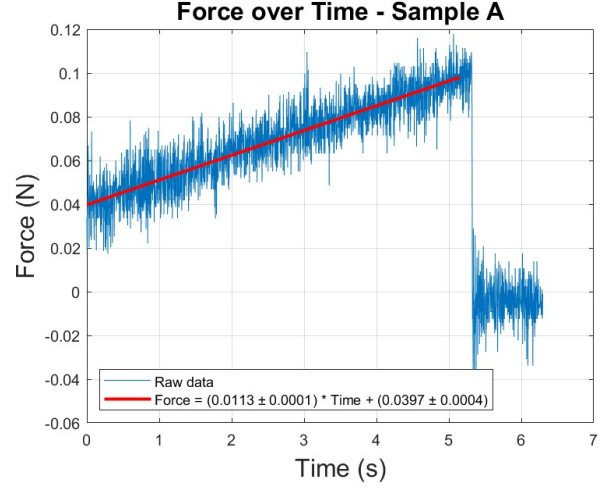


FIG. 8: Plot of the pulling force (in Newton) during tensile test over time (in second) for Sample A. Equation of the linear regression for sample A is $F_A = (0.0113 \pm 0.0001) \cdot t + (0.0397 \pm 0.0004)$. The force drop illustrate the total break of the sample.

Following the method of the linear regression for each sample, the linear expression of the force are listed below. The standard error are also specified, and are in the order of 10^{-2} lower than the slope coefficients of the linear regression.

$$\begin{aligned}
 F_A &= (0.0113 \pm 0.0001) \cdot t + (0.0397 \pm 0.0004) \\
 F_B &= (-0.0010 \pm 0.0004) \cdot t + (0.0180 \pm 0.0006) \\
 F_C &= (0.0120 \pm 0.0005) \cdot t + (0.0004 \pm 0.0006)
 \end{aligned}$$

The force at the moment of crack propagation is estimated for all three samples (Table II) :

Sample	A	B	C
Breaking time (s)	5.300	0.174	2.235
(Force $\pm$ 0.001) (N)	0.100	0.017	0.028

TABLE II: Force exerted on each sample before crack propagation. The uncertainties are computed using the propagation of uncertainties in Eq. B4.

The cross-section area A_C are measured with the electronic caliper and are listed in Table III.

Sample	A	B	C
$A_C (\text{mm}^2)$	(7.34 ± 0.19)	(4.66 ± 0.13)	(4.34 ± 0.14)

TABLE III: Cross section area of the samples. The uncertainties are computed following equation B2.

Regarding the crack length, the measurements obtained are listed in Table IV.

Lastly, every variables being calculated, the stress intensity factor comes naturally and we obtain the values given in Table V.

Sample	A	B	C
Crack length (mm)	3.958	6.458	4.627

TABLE IV: Measured crack lengths (in mm) for each sample

Sample	K_{linear} ($N \cdot m^{3/2}$)
A	(1695.04 ± 45.52)
B	(545.04 ± 15.72)
C	(756.87 ± 39.05)

TABLE V: Intensity stress factor in linear assumption for each sample. The uncertainties are computed following Eq. B7.

Discussion.—In this study, we compared two different approaches of calculating the stress intensity factor (K): the energy-based formula (Eq. 6) $K_{plast} = \sqrt{J \cdot \tilde{E}}$, suitable for the plastic behavior, and the linear elastic formula (Eq. 11): $K_{lin} = 1.12\sigma\sqrt{\pi a}$ assuming linear behavior. The results obtained are summarized in the Table VI and the relative errors between the two methods were quantified as 0.1%, 2.3% and 33.9%.

Sample	A	B	C
K_{linear} ($N \cdot m^{3/2}$)	1695	545	756
$K_{Plastic}$ ($N \cdot m^{3/2}$)	1704	536	1317
Relative error (%)	0.5	1.6	42.6

TABLE VI: Comparison of the intensity stress factor between linear and plastic behaviors for each sample, along with the corresponding relative error.

The stress intensity factor in both methods exhibits an excellent agreement for the first two samples. The discrepancies are within the acceptable ranges for many engineering applications and suggest that the linear assumption provides a sufficiently accurate representation of the stress intensity factor, even in the presence of minor plasticity. The relative error for sample 3 becomes significantly larger up to 42.6%. This deviation seems to indicate that the linear behavior does not adequately describe the fracture of the gel sample. The large error suggests that the material exhibits strong plastic behavior during fracture. The formula of $K_{Plastic}$ seems more appropriate and yields a more accurate representation of the material's fracture.

However, this hypothesis can be ruled out, as Sample C is made from the same material as the other samples. There could be several other reasons why the final sample does not conform to the linear hypothesis. While the experimental conditions were kept as consistent as possible across tests, it is possible that the last sample may have

slightly dried out while waiting for Tests A and B to be completed. Since humidity significantly affects the material's properties, as noted by Subramani *et al.* (2020)[10], such a change could account for the observed behavior. In the future, conducting additional tests in a more controlled environment, especially regarding gels' moisture could provide more accurate and reliable results. These new tests will help determine whether sample C should be considered or if it is an anomaly.

This study investigated the fracture behavior of polyacrylamide hydrogels using Digital Image Correlation (DIC) to measure deformation fields near crack tips. By calculating stress intensity factors under both linear elastic and plastic assumptions, we assessed the material's response to applied stress. Our results showed that for most samples, the linear elastic model provided a reliable estimate of the stress intensity factor, with minimal error. However, in cases where plastic deformation was more pronounced, the plastic stress intensity factor ($K_{plastic}$) yielded more accurate results. The significant deviation observed in one sample (Sample C) further emphasized the importance of considering plasticity in fracture analysis. Overall, our findings demonstrate the effectiveness of DIC in characterizing crack-tip behavior and highlight the need for careful consideration of material nonlinearity in fracture mechanics, especially for soft materials like hydrogels.

In their studies, Wang *et al.* [1] and Lin *et al.* [11] investigated the mechanical properties of hydrogels at varying crosslinker-to-monomer ratios (C). They identified key parameters, such as polymer concentration and crosslinker stoichiometry that help assess and quantify hydrogel stiffness and other mechanical behaviors for given network architectures. As C increases, the network becomes more tightly crosslinked, reducing chain flexibility and increasing stiffness, resulting in a higher Young's modulus (E). Future studies could focus on identifying groups of crosslinker-to-monomer ratio (C) values or specific network architectures that result in brittle behavior, as well as those that lead to ductile behavior. Such research would enhance our understanding of crack propagation, allowing for more accurate predictions of when linear approximation models are valid for evaluating material failure.

CRedit Author Statement¹

Thomas Lepère: Software, Data Curation, Formal analysis, Writing - Review Editing. **Paul Nauche:** Conceptualization, Methodology, Visualization, Validation, Writing - Original Draft. **Théophile de Tournemire:** Conceptualization, Writing - Original Draft, Writing - Review Editing, Formal analysis.

¹ The redaction of the text has been AI-assisted

Appendix A: Force over time for samples B and C

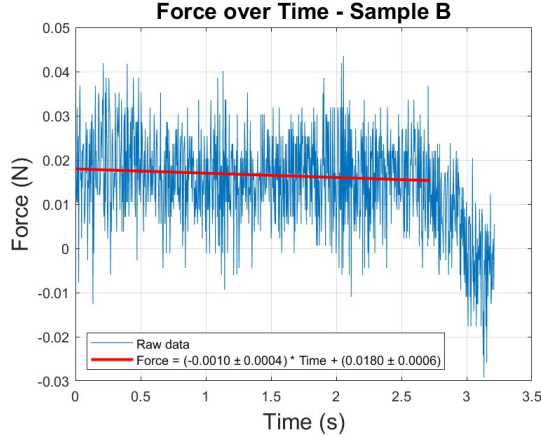


FIG. 9: Exerted force (in N) on sample B over time (in s). Linear regression expression $F_B = (-0.0010 \pm 0.0004) \cdot t + (0.0180 \pm 0.0006)$. The linear expression indicates a nearly constant force throughout the test, attributed to the propagation of the crack. The complete failure of the sample occurs when the force abruptly drops at the end.

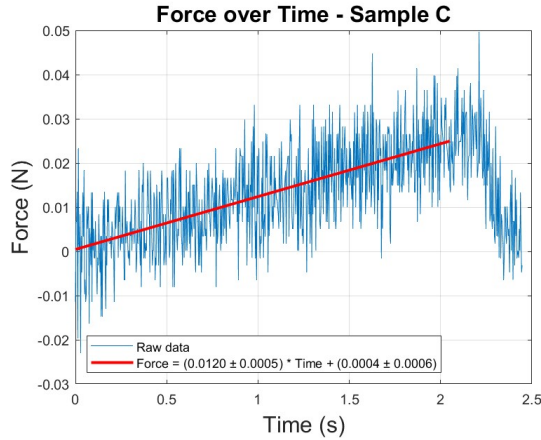


FIG. 10: Exerted force (in N) on sample C over time (in s). Linear regression expression $F_C = (0.0120 \pm 0.0005) \cdot t + (0.0004 \pm 0.0006)$

Regarding sample B, an increasing force slope is expected as the sample stretches. Instead, the force is constant. This is explained by the fact that the sample broke really fast, within the eight first frames of the video. This is not enough to observe a change in the force data. The sample is still taken into account in this study, as variations in results and conditions are part of the scientific process.

Appendix B: Uncertainties

With the uncertainties for the width and thickness equals ($\Delta W = \Delta T = \Delta x$), the uncertainty for the cross-section area can be computed as :

$$A_C = W \cdot t \quad (B1)$$

$$\Delta A_C = \sqrt{W^2 + T^2} \cdot \Delta x \quad (B2)$$

For the breaking force obtained in the Table 11, the propagation of uncertainties is used :

$$F = a \cdot F + b \quad (B3)$$

$$\Delta F = \sqrt{\left(\frac{\partial F}{\partial t} \cdot \Delta t\right)^2 + \left(\frac{\partial F}{\partial a} \cdot \Delta a\right)^2 + \left(\frac{\partial F}{\partial b} \cdot \Delta b\right)^2} = t \cdot \Delta a \quad (B4)$$

The uncertainty of the linear stress intensity factor is computed as follow :

$$K = 1.12\sigma\sqrt{\pi a} = 1.12\frac{F}{A_C}\sqrt{\pi a} \quad (B5)$$

$$\Delta K = \sqrt{\left(\frac{\partial F}{\partial F} \cdot \Delta F\right)^2 + \left(\frac{\partial A_C}{\partial A} \cdot \Delta A_C\right)^2 + \left(\frac{\partial F}{\partial a} \cdot \Delta a\right)^2} \quad (B6)$$

$$= \frac{1.12\sqrt{\pi a}}{A_C} \cdot \sqrt{(\Delta F)^2 + \left(\frac{-F}{A} \cdot \Delta A_C\right)^2} \quad (B7)$$

Appendix C: Piola-Kirchoff stress tensors for samples B and C

P_{22} of the Piola-Kirchoff stress tensor

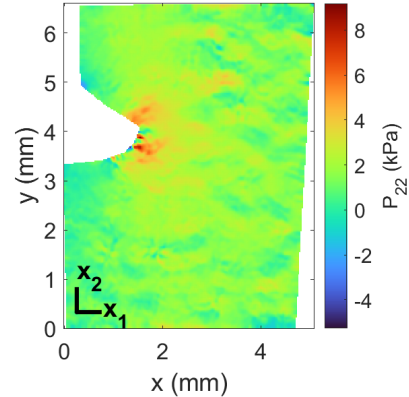


FIG. 11: Second component of the first Piola-Kirchoff stress tensor for sample B.

P_{22} of the Piola-Kirchoff stress tensor

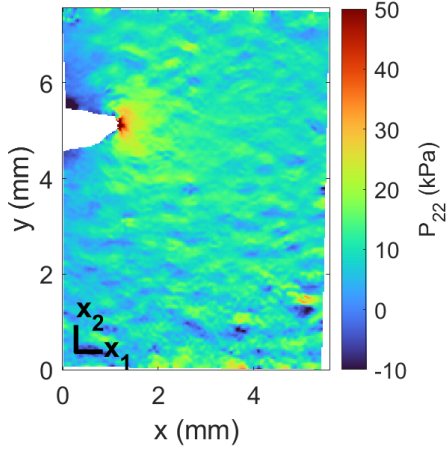


FIG. 12: Second component of the first Piola-Kirchoff stress tensor for sample C.

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